

Linear Transformations

Geometric Algorithms
Lecture 7

Practice Problem

Find three vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^3$ such that

- » every pair of vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$, $\{\mathbf{v}_1, \mathbf{v}_3\}$, $\{\mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent
- » $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent

Objectives

- » Introduce matrix transformations
- » Define linear transformations
- » Start looking at the geometry of linear transformations

Keywords

Transformations

Domain, Codomain

Image, Range

Matrix Transformations

Linear Transformations

Additivity, Homogeneity

Dilation, Contraction, Shearing, Rotation

Matrix Transformations

Recall: Spans (with Matrices)

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Df. The *span* of a set of vectors is the set of all possible linear combinations

$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \{[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{v} : \mathbf{v} \in \mathbb{R}^n\}$$

Matrices as Transformations

Matrices as Transformations

Matrices *transform* vectors into vectors the span of its columns

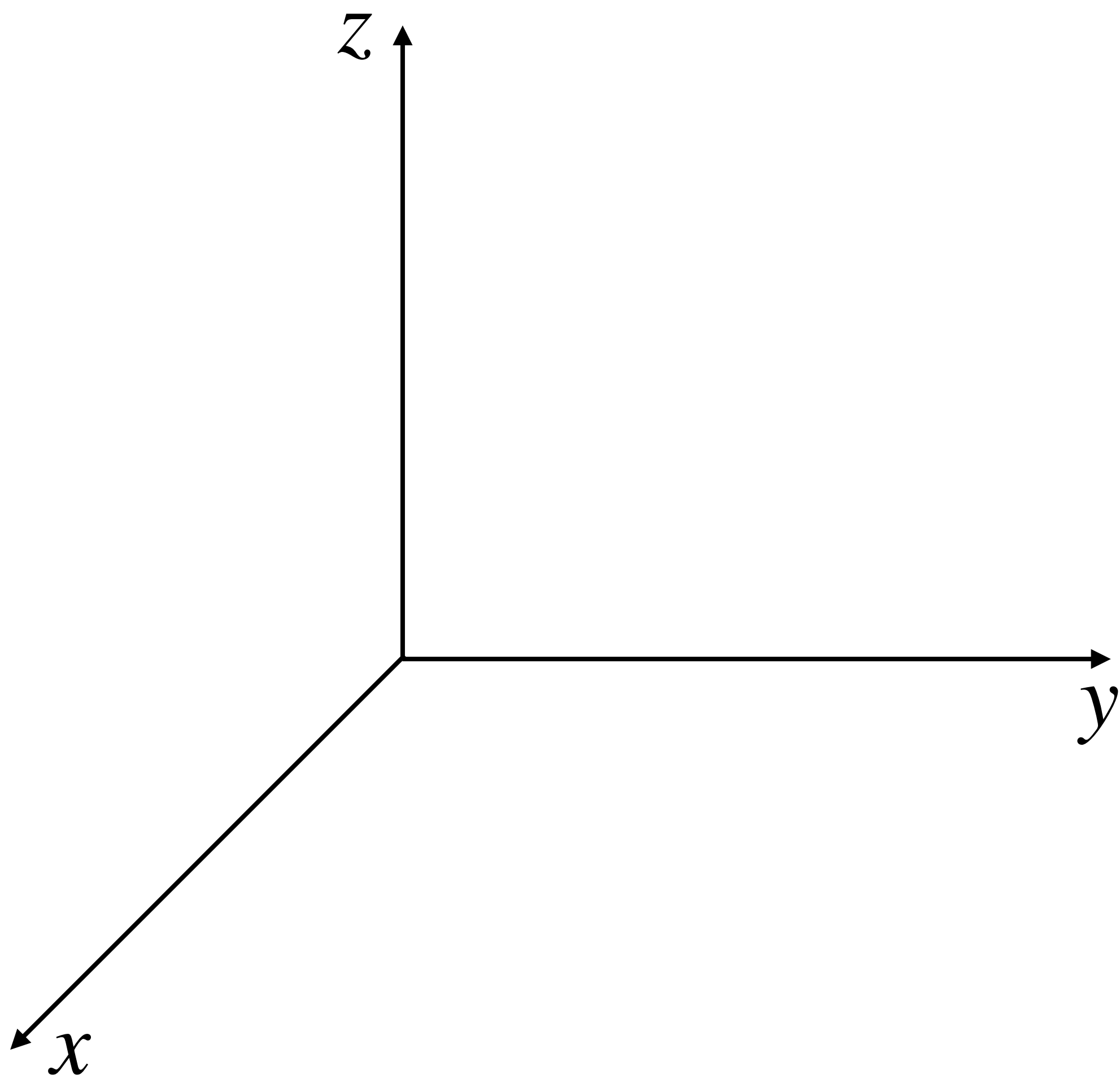
$$\mathbf{x} \mapsto A\mathbf{x}$$

map a vector $\mathbf{v}$ to the vector $A\mathbf{v}$

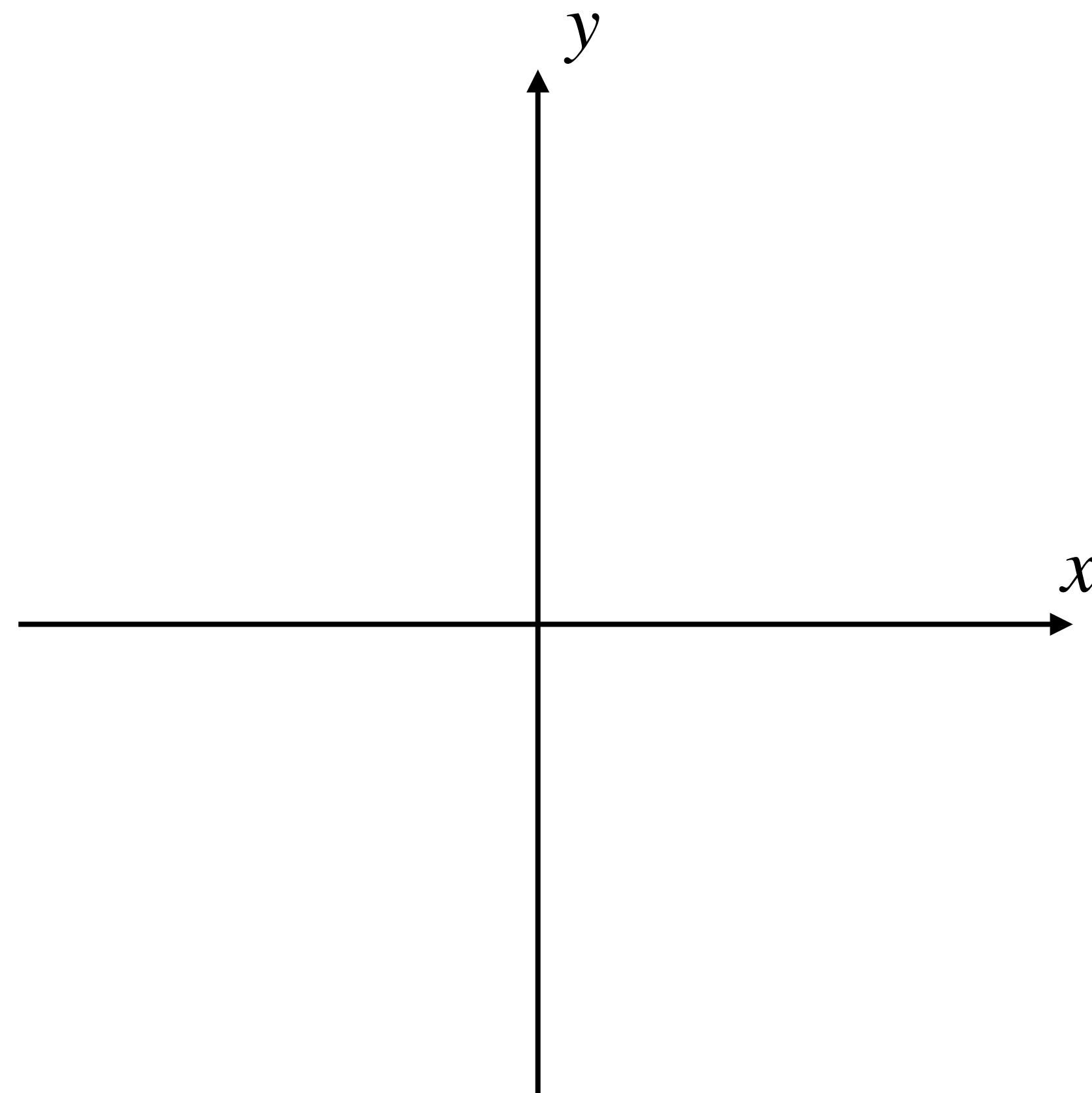
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} =$$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \mathbf{x}$$

Important

The vector may be a different
size after translation

Recall: Matrix-Vector Multiplication

matrix-vector multiplication only works if the number of *columns* of the matrix matches the dimension of the vector

$$\begin{array}{c} \textcolor{blue}{m} \left[\begin{array}{ccc} * & \dots & * \\ * & \dots & * \\ \vdots & \ddots & \vdots \\ * & \dots & * \\ * & \dots & * \end{array} \right] \quad \textcolor{red}{n} \left[\begin{array}{c} * \\ \vdots \\ * \end{array} \right] = \textcolor{blue}{m} \left[\begin{array}{c} * \\ * \\ \vdots \\ * \\ * \end{array} \right] \\ (m \times n) \quad \mathbb{R}^n \quad \mathbb{R}^m \end{array}$$

Motivating Questions

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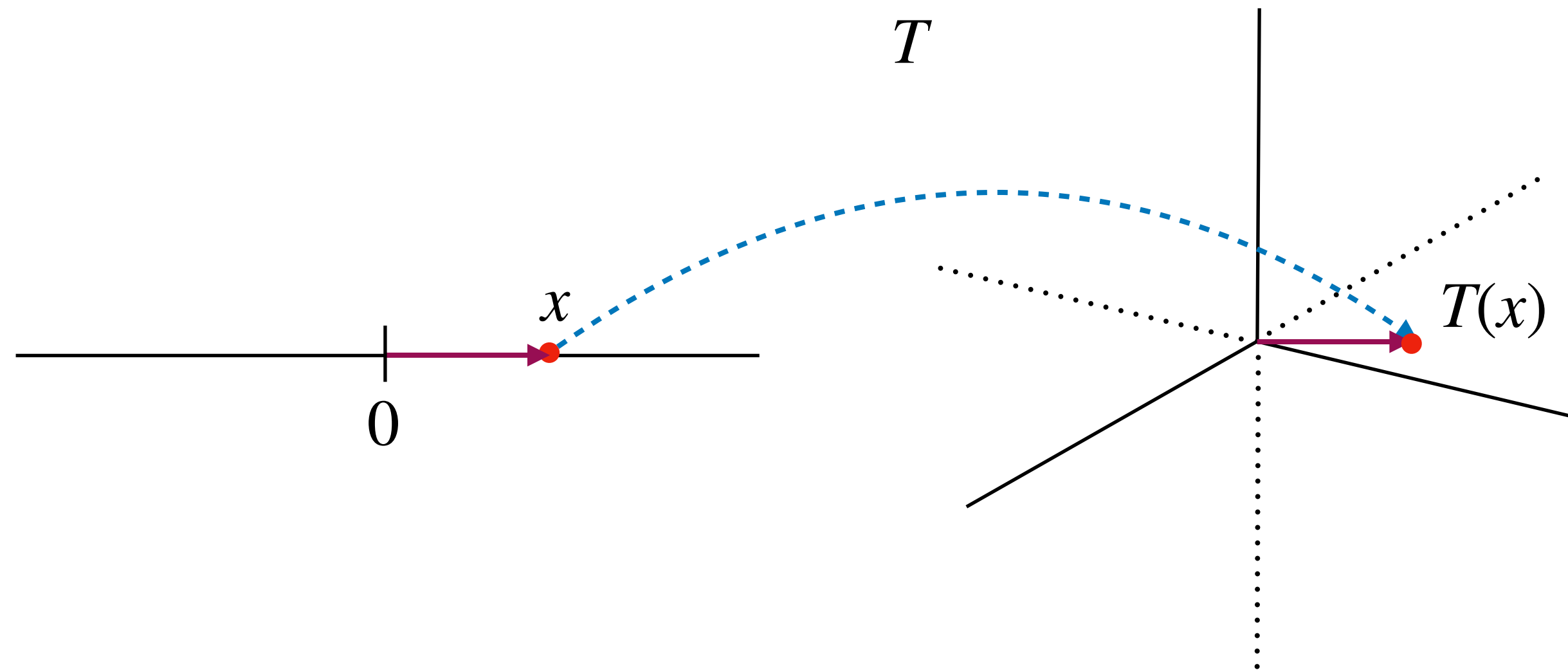
New Interpretation

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A
transforms into $\mathbf{b}$?

Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A
transforms into $\mathbf{b}$

Question

Suppose a matrix transforms vectors according to the following picture. *What's the shape of the matrix?*



$$\mathbb{R}^n \rightarrow \mathbb{R}^n$$

Mapping between the same space can be viewed
as a way of "moving around points"

Transformations

Transformations in General

Df. A *transformation* T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function which maps every vector $\mathbf{v}$ in $\mathbb{R}^n$ to a vector $T(\mathbf{v})$ in $\mathbb{R}^m$

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$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

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$$T : \underbrace{\mathbb{R}^n}_{\text{domain}} \rightarrow \underbrace{\mathbb{R}^m}_{\text{codomain}}$$

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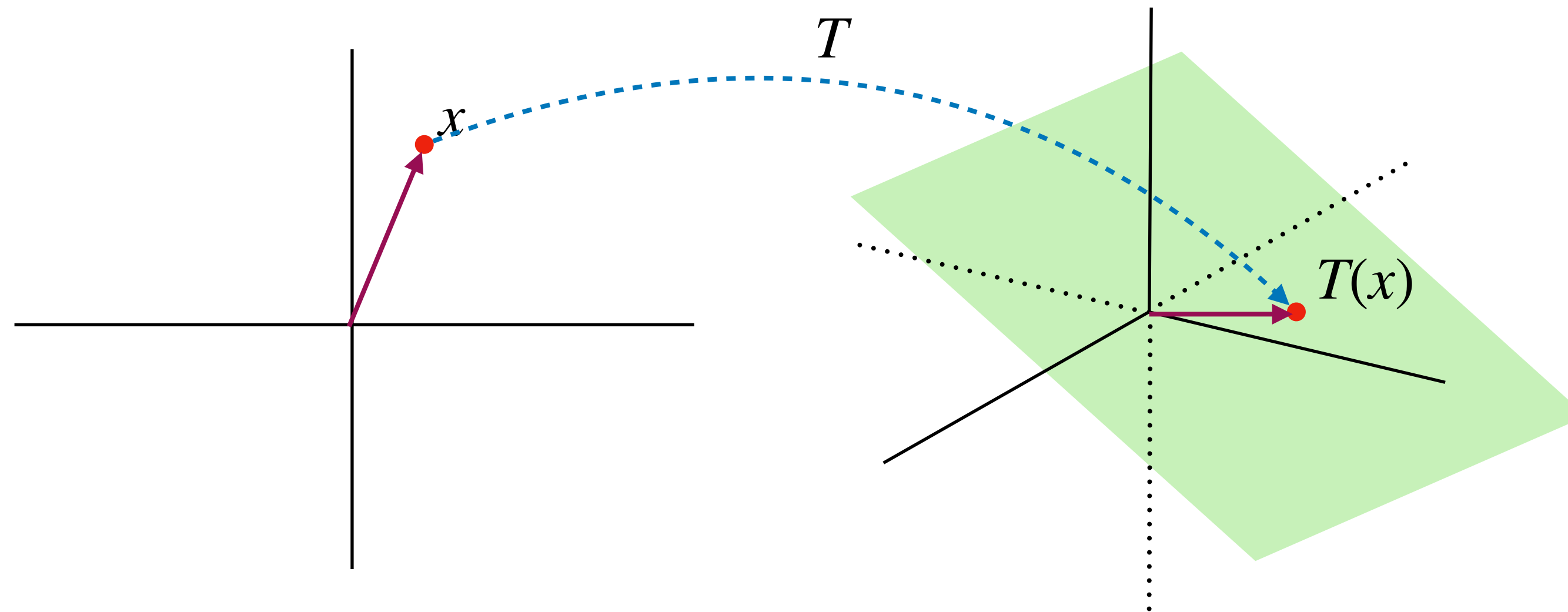
$$\text{ran}(T) = \{T(\mathbf{v}) : \mathbf{v} \in \mathbb{R}^n\}$$

image of $\mathbf{v}$ under $T \equiv$ output of T applied to $\mathbf{v}$

range of $T \equiv$ all possible output of T

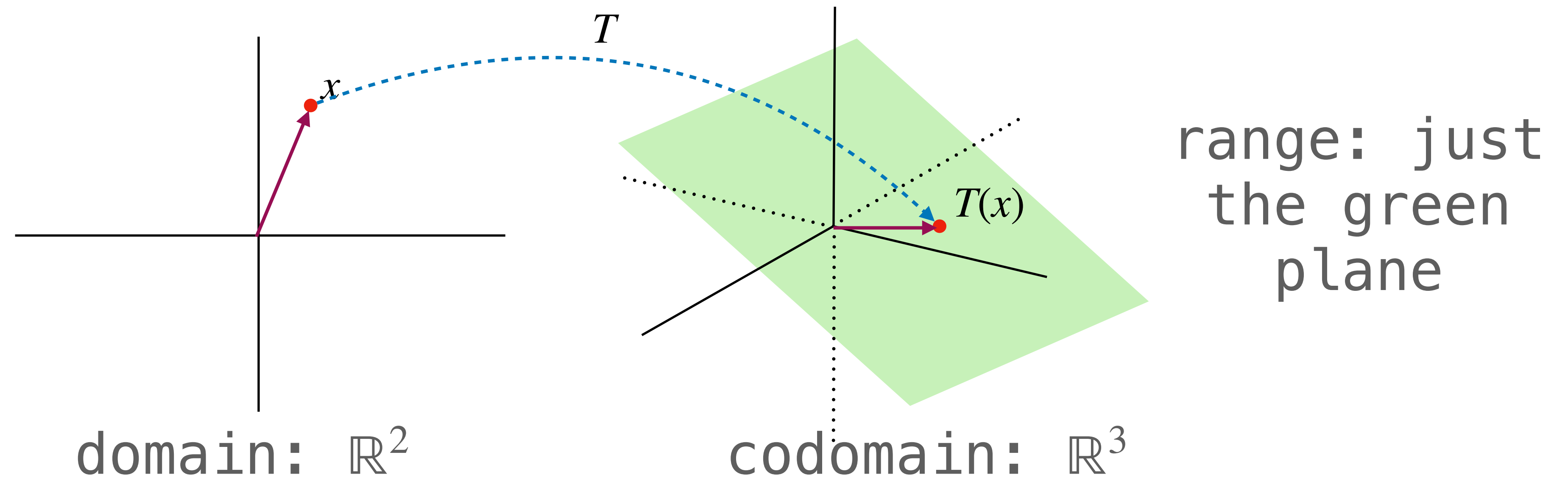
Codomain and Range

The codomain and range of a transformation may or may not be the same



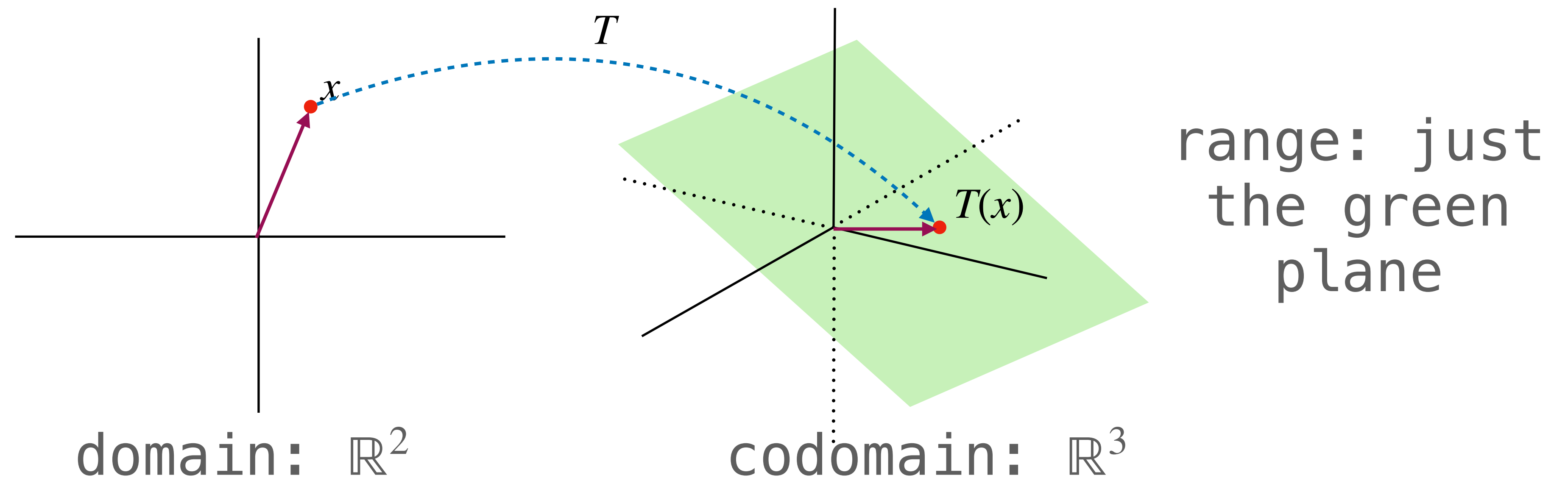
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The range is always contained in the codomain

Example

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1^2 \\ x_2 \\ 0 \end{bmatrix}$$

Matrix Transformations

Transformation of a Matrix

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Df. The *transformation of a* $(m \times n)$ *matrix* A is the function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

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$$T(\mathbf{v}) = A\mathbf{v}$$

i.e., Given $\mathbf{v}$, return A multiplied by $\mathbf{v}$

e.g., $T(\mathbf{v}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{v}$

Range and Span

$$\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \text{ran}([\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n])$$

Linear Transformations

Motivating Questions

What kind of functions can we define this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

Recall: Algebraic Properties of Matrix-Vector Multiplication

Recall: Algebraic Properties

Matrix-vector multiplication satisfies the following two properties:

1. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ (additivity)

2. $A(c\mathbf{v}) = c(A\mathbf{v})$ (homogeneity)

Example

Linear Transformations

Df. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Linear Transformations

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Matrix transformations are linear transformations

Example: Identity

$$T(\mathbf{v}) = \mathbf{v}$$

Example: Zero

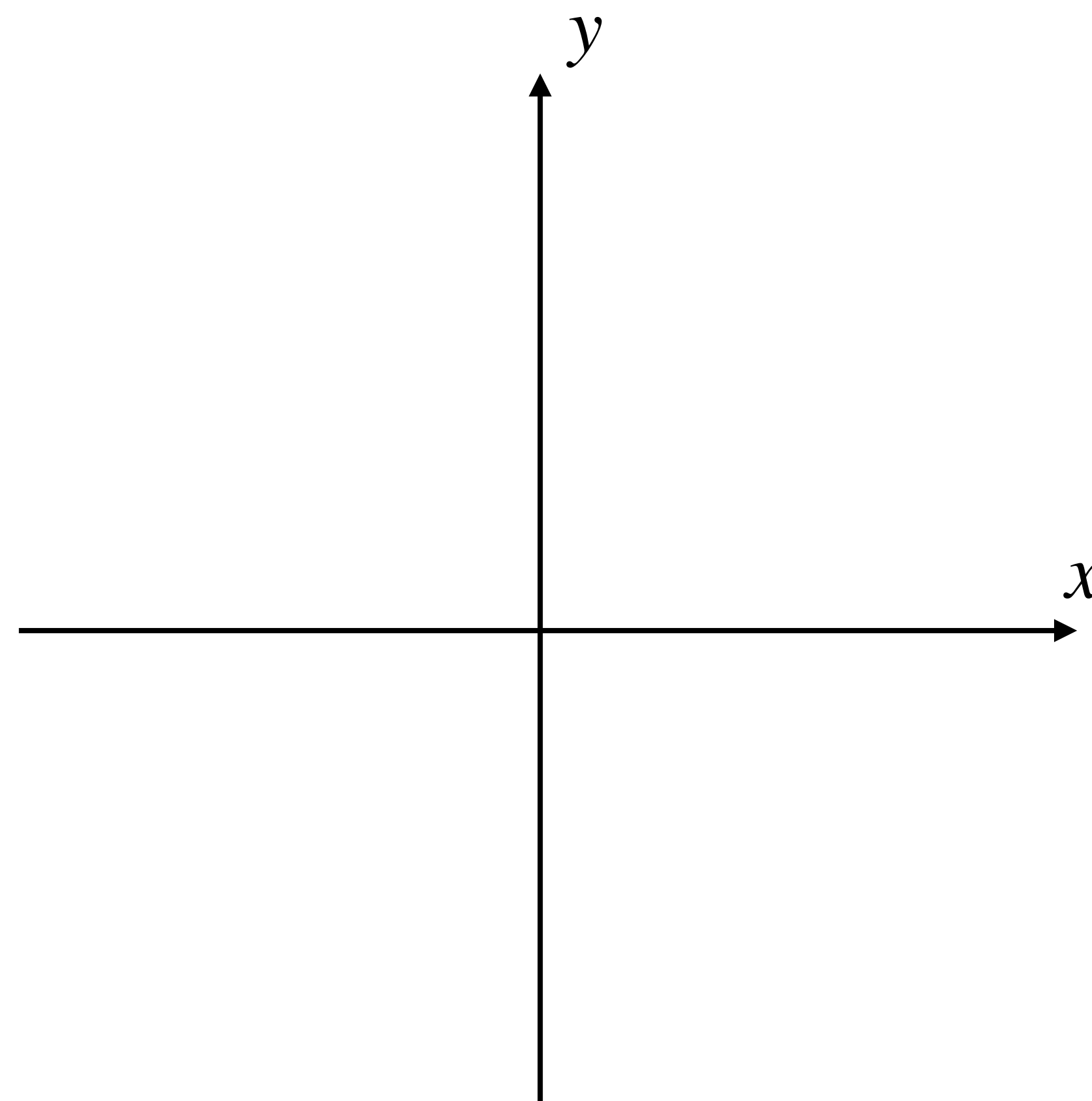
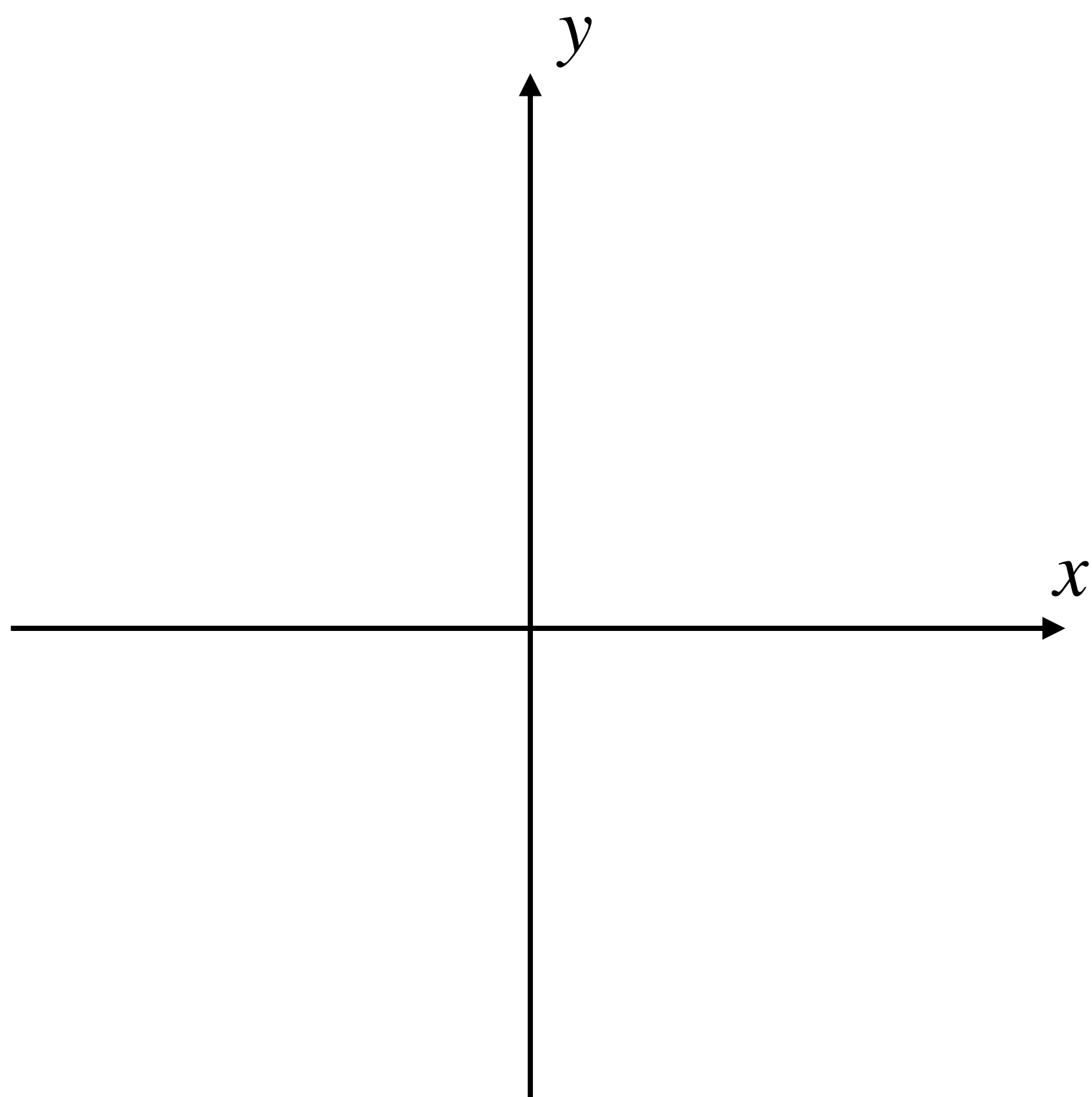
$$T(\mathbf{v}) = \mathbf{0}$$

Example

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + x_2 \\ x_2 - x_3 \\ 3x_1 \end{bmatrix}$$

Example: Rotation

We'll see this next time, but for now we can use pictures



Example: Indefinite Integrals

$$T(f) = \int f(x) dx$$

Disclaimer:
Advanced
Material

Example: Expectation

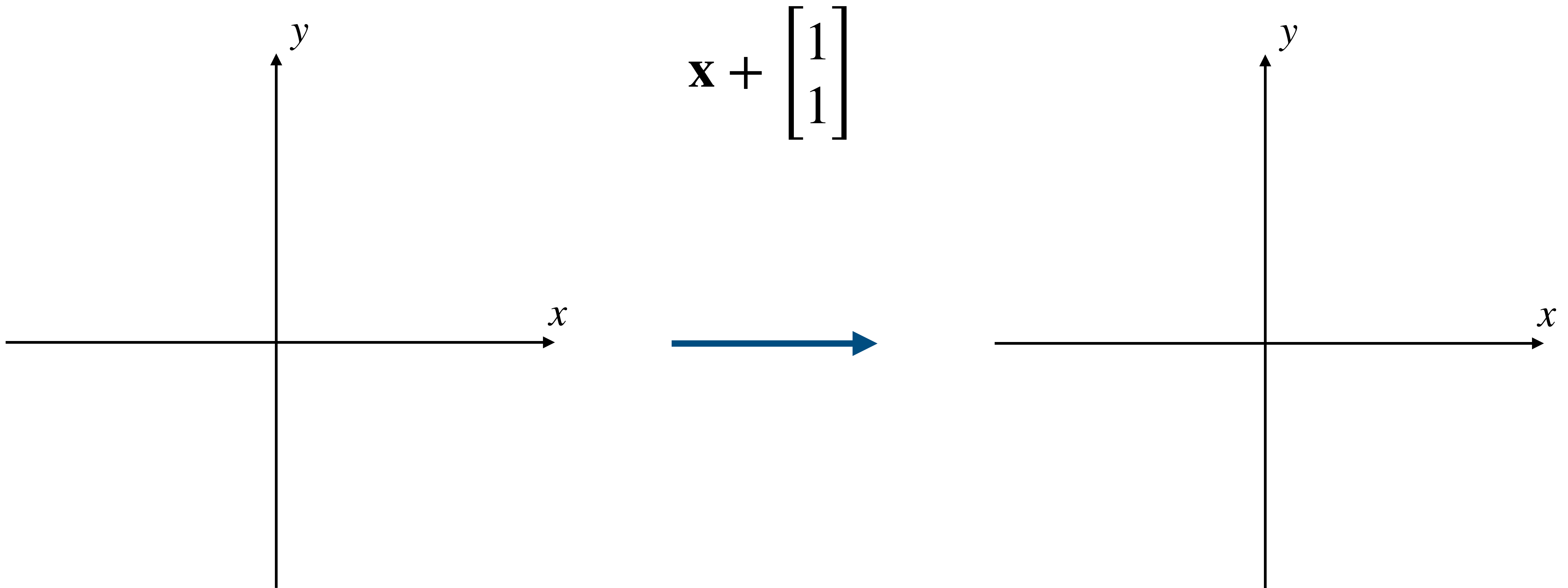
$$T(X) = \mathbb{E}[X]$$

Disclaimer:
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Material

Non-Example: Squares

$$T(x) = x^2$$

Non-Example: Translation



Properties of Linear Transformations

The Zero Vector

The Zero Vector

$$T(\mathbf{0}) = ???$$

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$

The zero vector is *fixed* by linear transformations

The Zero Vector

$$T(\mathbf{0}) = \mathbf{0}$$
Two red arrows originate from the text 'Note: These may be different dimensions!' below. One arrow points diagonally up and to the left towards the bold zero vector inside the parentheses of the equation $T(\mathbf{0}) = \mathbf{0}$. The other arrow points diagonally up and to the right towards the bold zero vector on the right side of the equation.

Note: These may be different dimensions!

The zero vector is *fixed* by linear transformations

A Single Condition

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$$T(a\mathbf{v} + b\mathbf{u}) = aT(\mathbf{v}) + bT(\mathbf{u})$$

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We can combine our linearity conditions

A Single Condition

Thm. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is linear if and only if for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^m$ and any real numbers a and b ,

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

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It's often easiest to show this
single condition

Linear Combinations

Geometry of Matrix Transformations

Motivating Questions

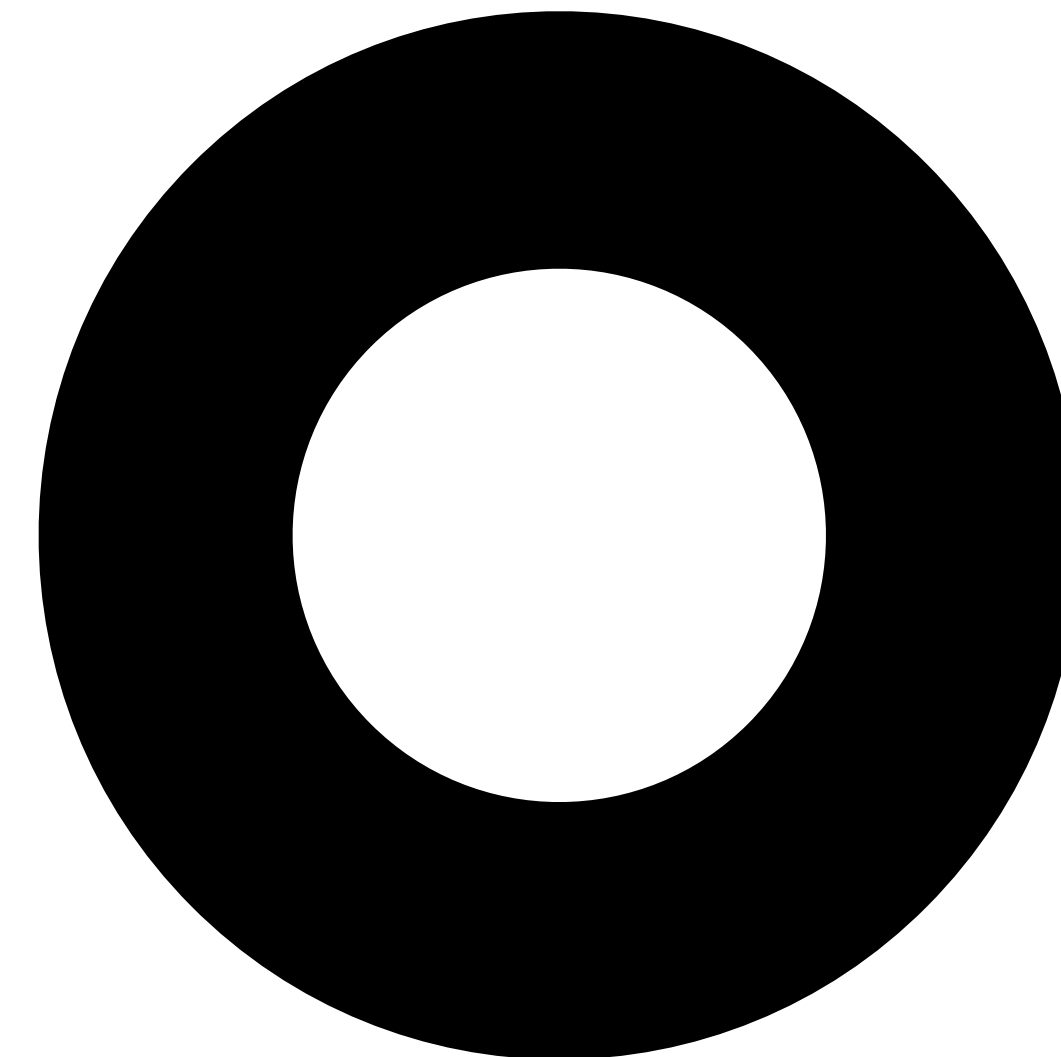
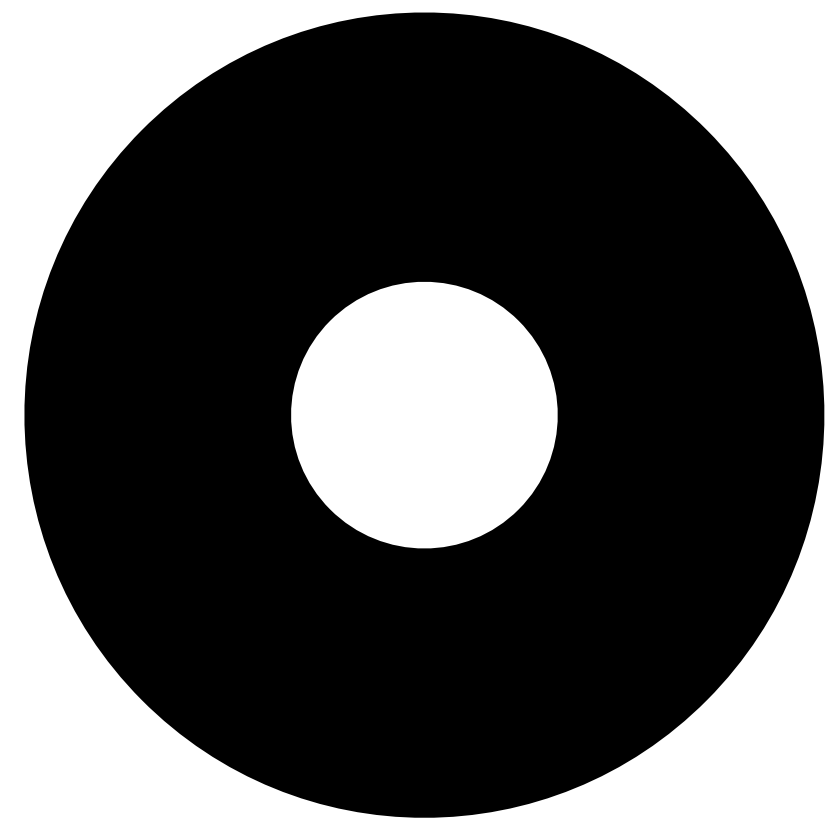
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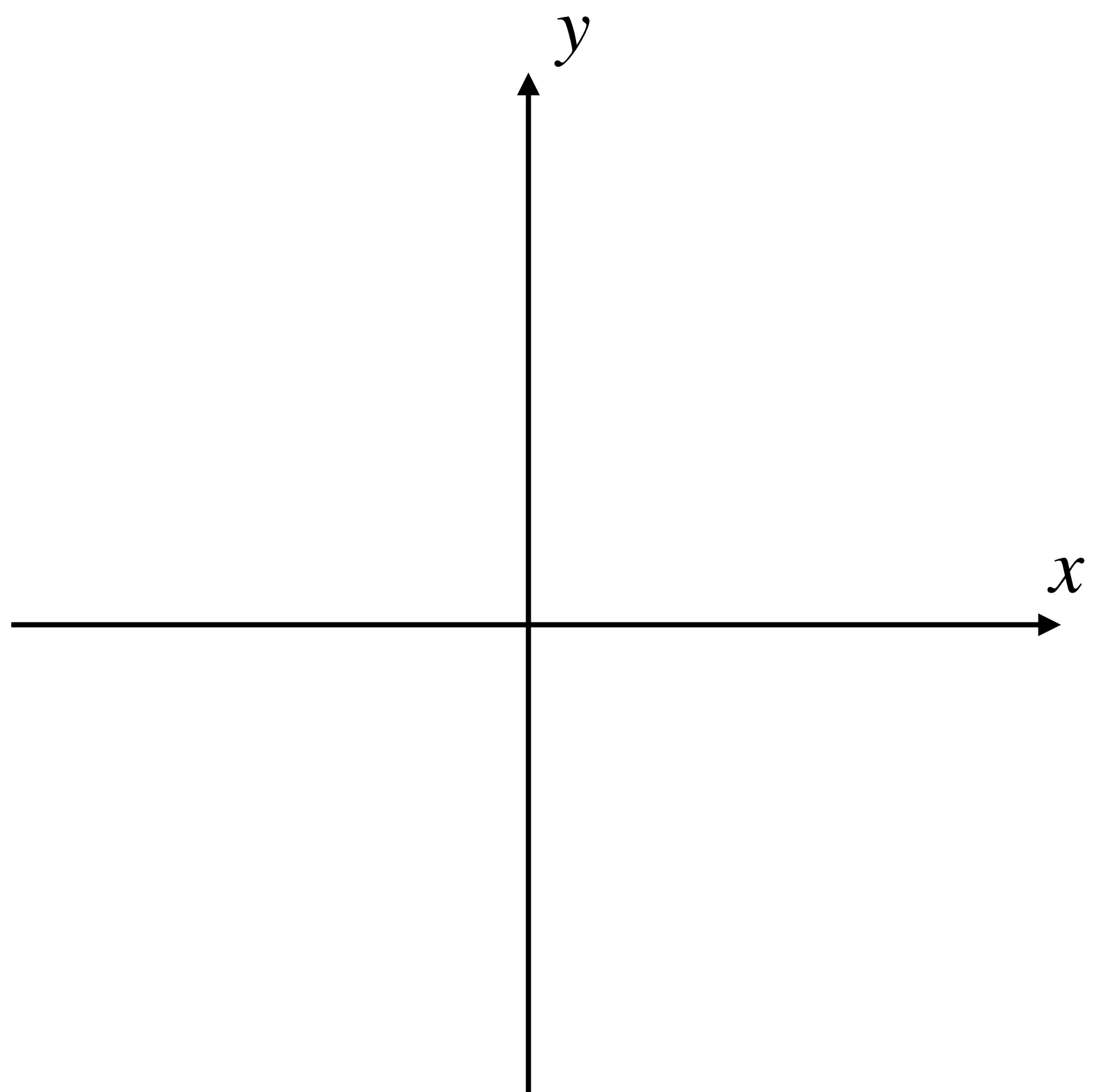
Matrix transformations change the
"shape" of a set of set of
vectors (points)

Example: Dilation

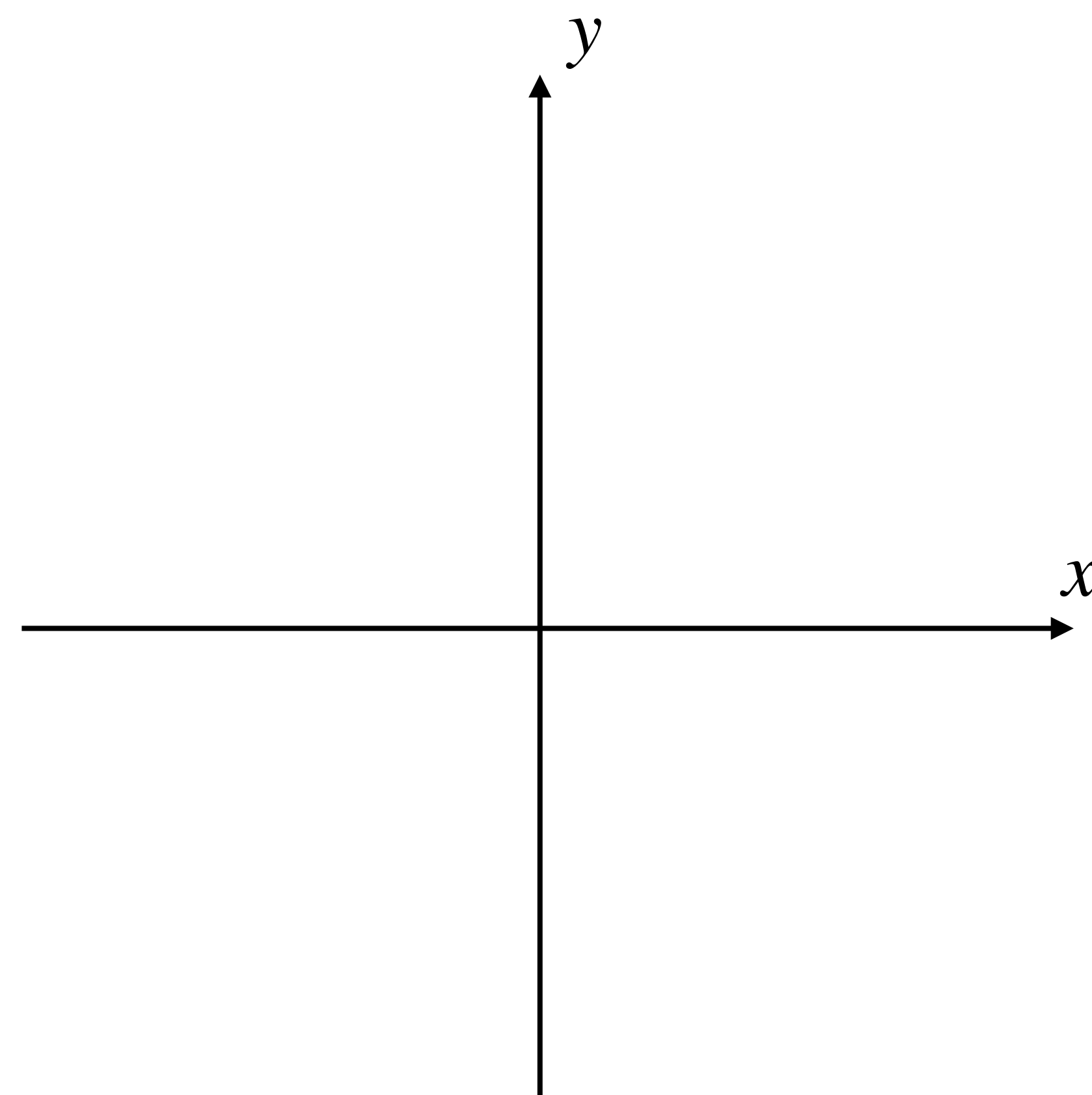


Example: Dilation

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$

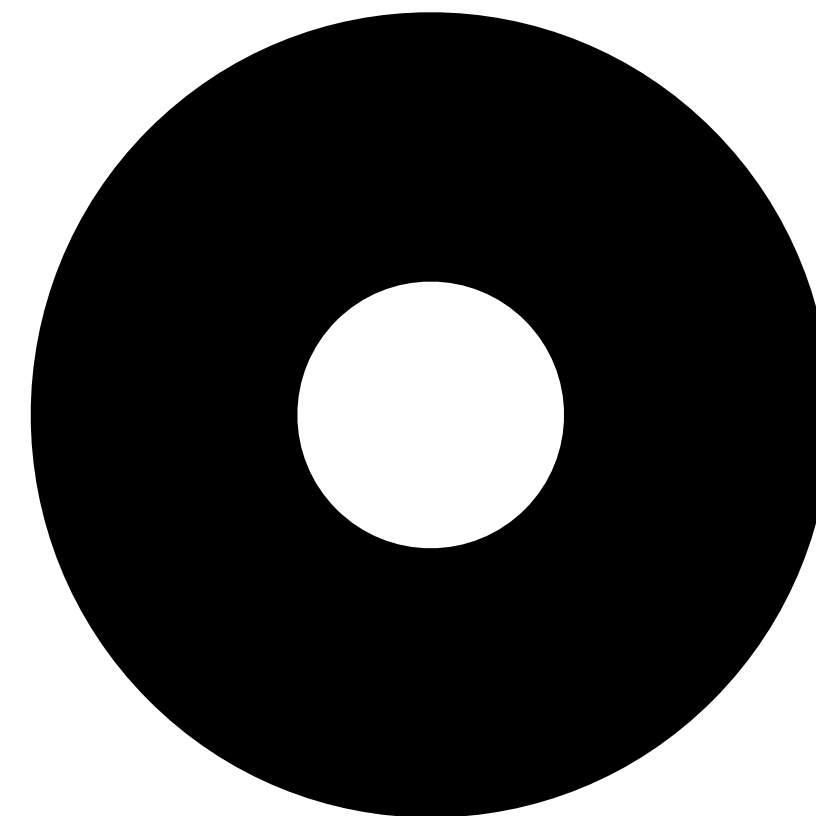
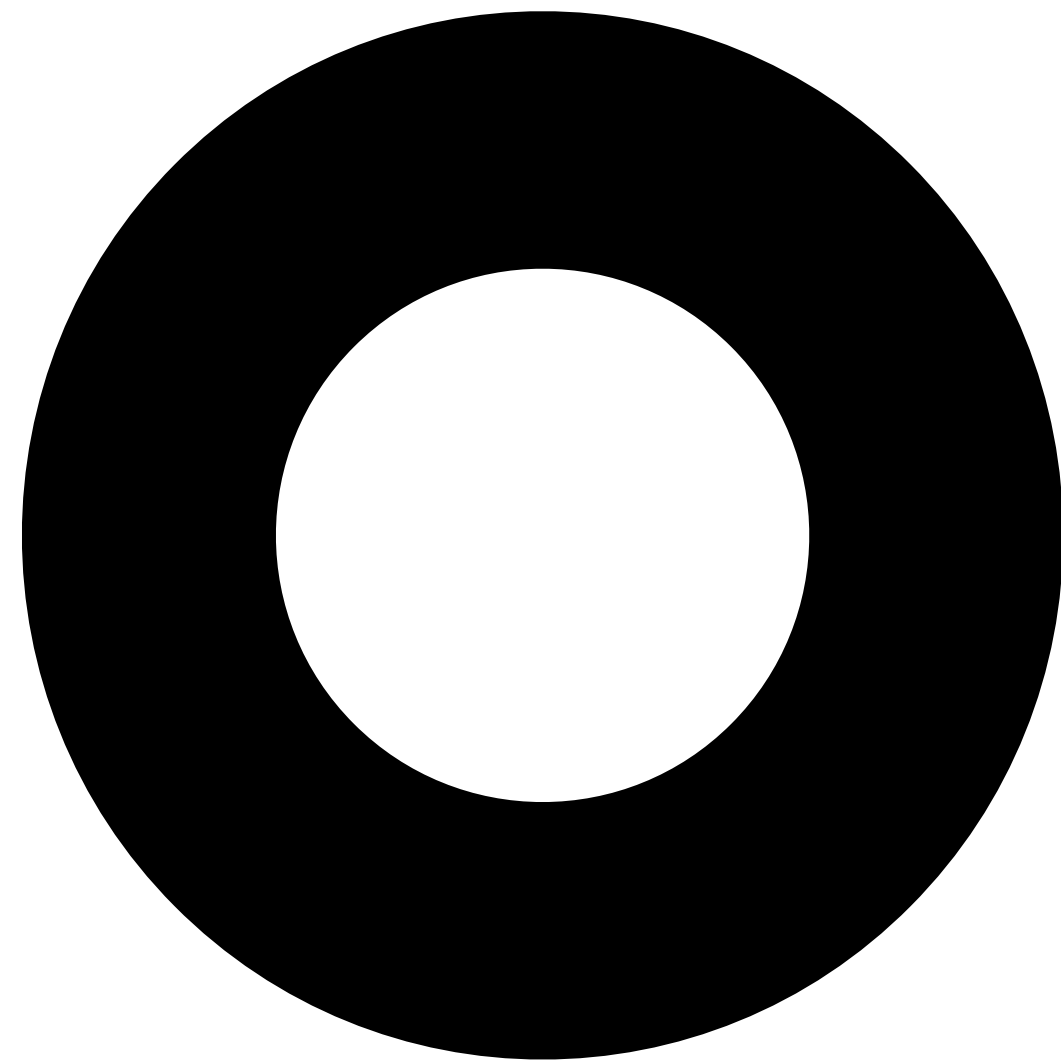


$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}$$

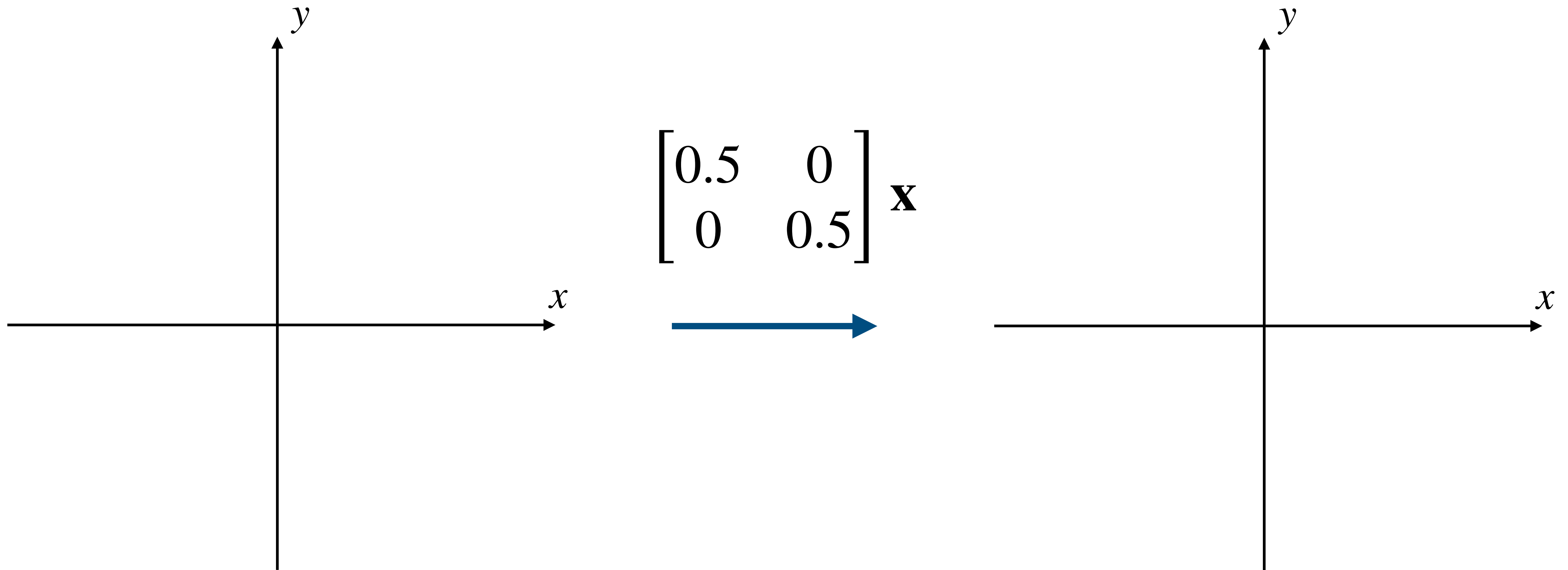
if $r > 1$, then the transformation pushes points away from the origin

Example: Contraction



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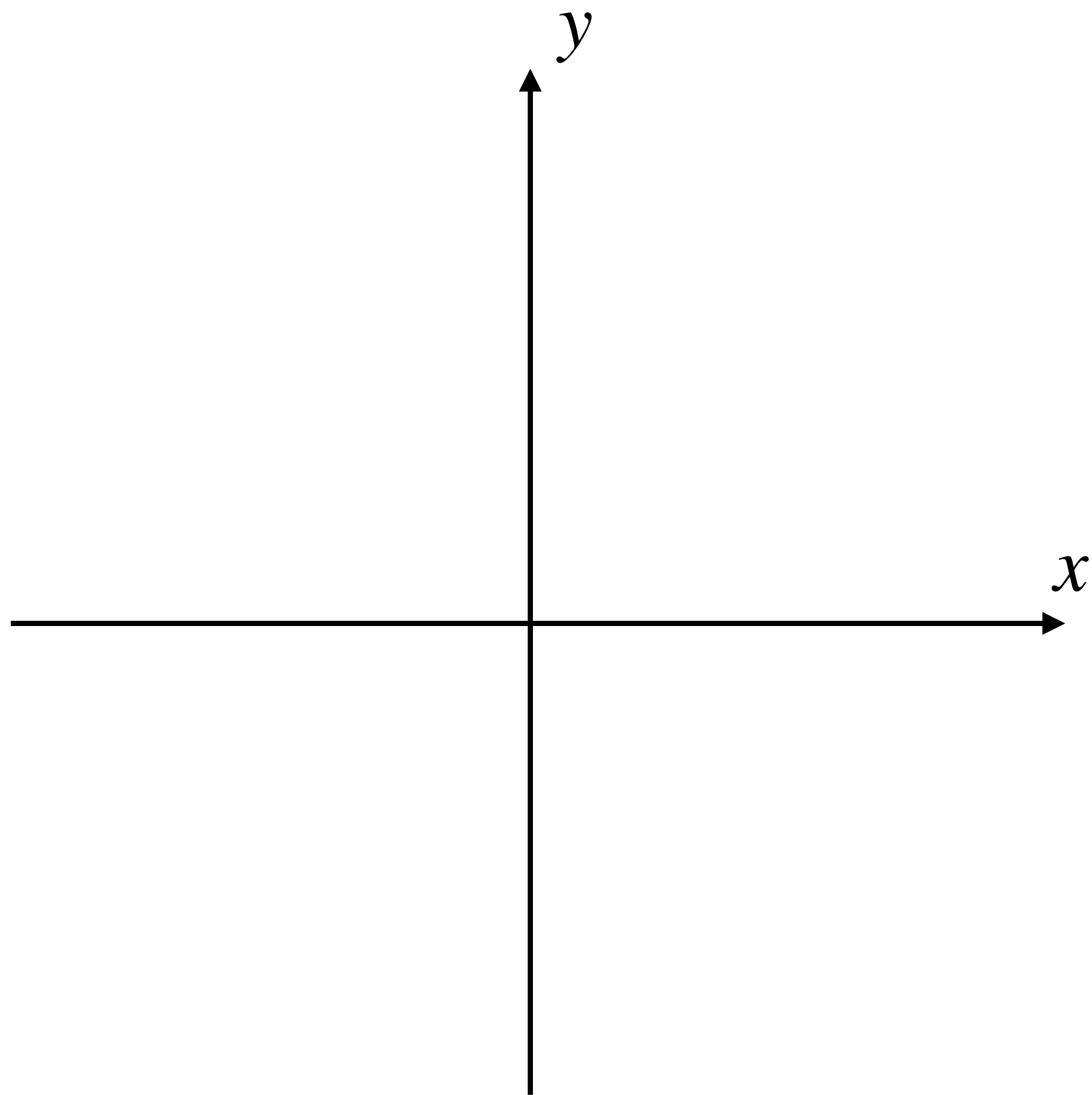
if $0 \leq r \leq 1$, then the transformation pulls points towards the origin

Example: Shearing

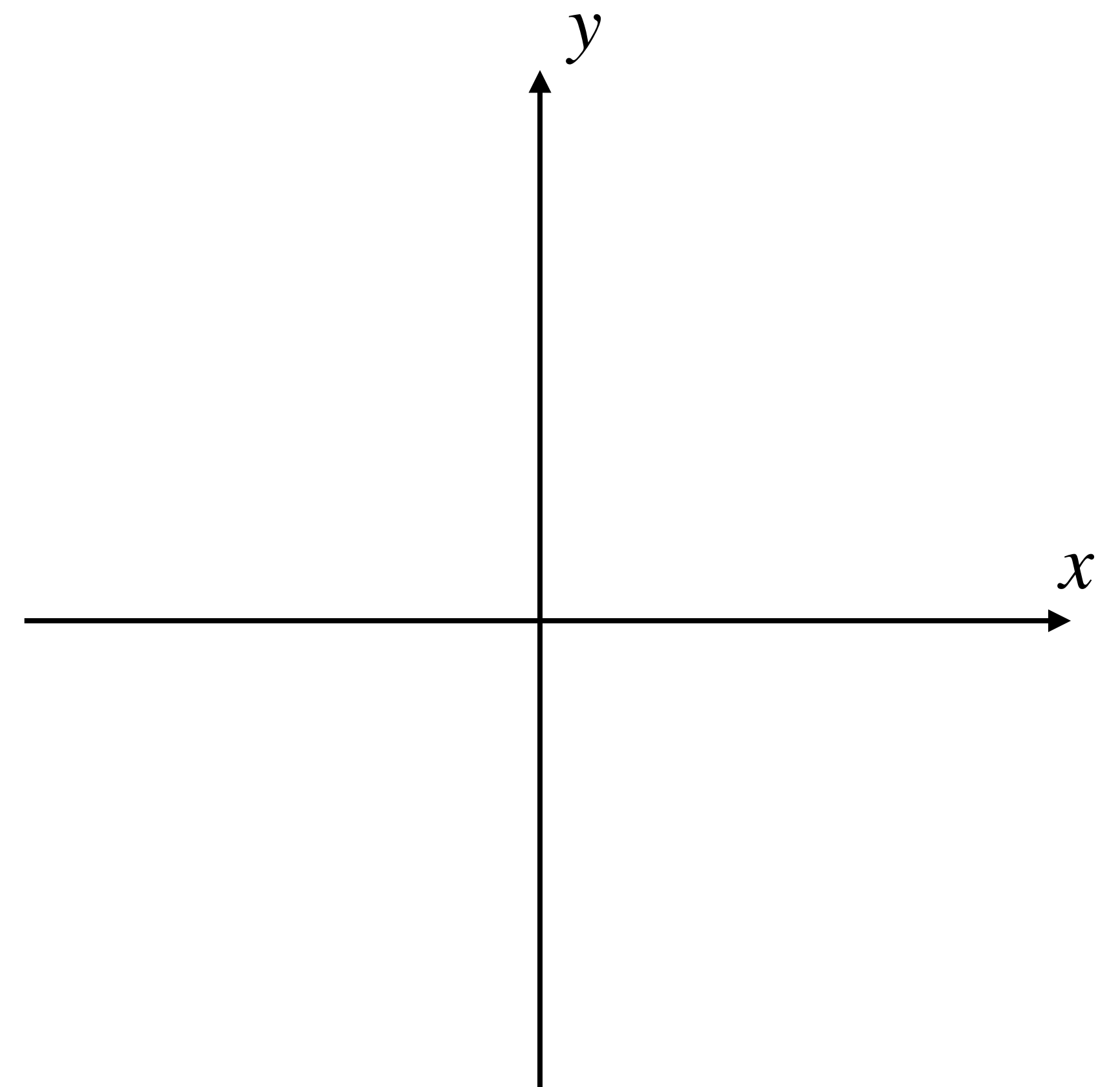


Example: Shearing

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$

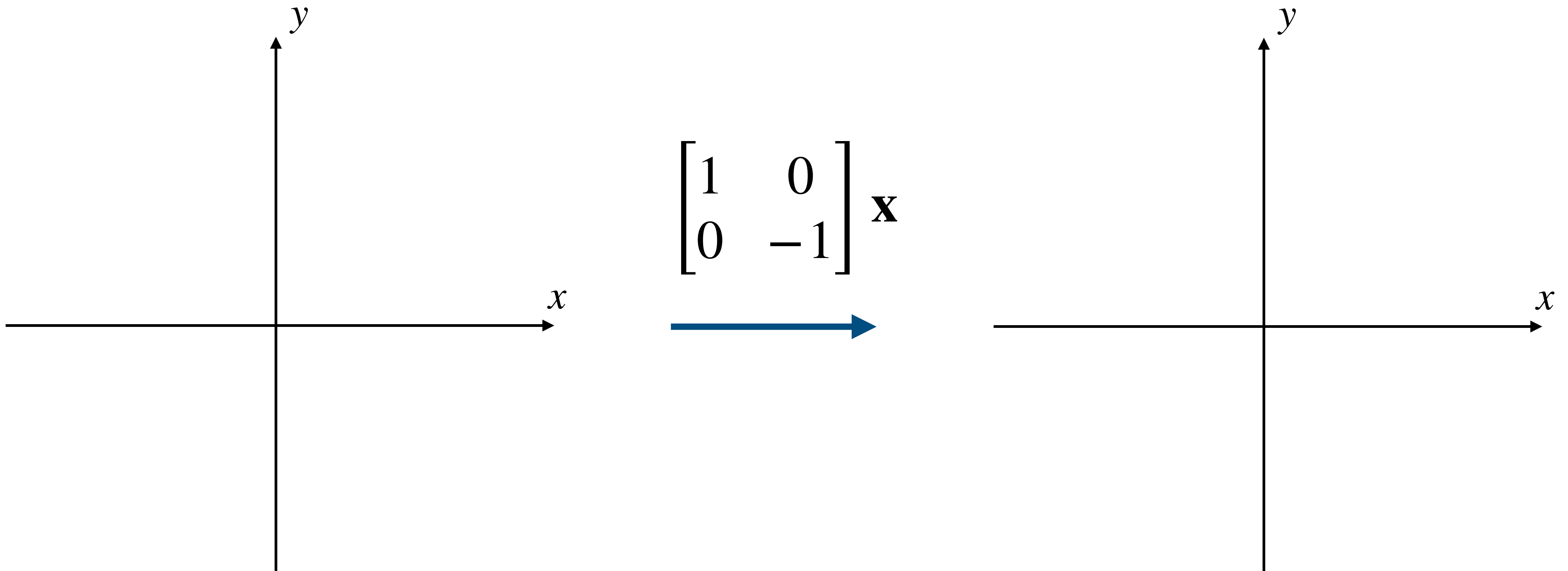



Question

$$\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$

Draw how this matrix transforms points. What kind of transformation does it represent?

Answer: Reflection



demo

Summary

Matrices can be viewed as **linear transformations**

Matrix transformations change the **shape** of points sets

Linear transformations behave well with respect to **linear combinations**